DECOR+: Energy-Balanced Representation Enhancement for Long-Tailed Remote Sensing Scene Recognition

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***Abstract*—Remote sensing (RS) imagery, characterized by wide geographic coverage and highly diverse land-cover categories, naturally exhibits severe long-tailed distributions. Such an imbalance causes head classes to dominate both feature learning and gradient optimization, leading to degraded separability, overlapping feature distributions, and biased decision boundaries. We interpret these effects through a unified energy imbalance perspective, where head classes accumulate disproportionate spectral, geometric, and statistical representation energy, while tail classes remain weakly expressed. To address this issue, we propose DECOR+ (energy-balanced representation enhancement), a unified framework that restores multidomain energy equilibrium for long-tailed RS scene recognition. DECOR+ consists of three complementary components. First, a spectral balancing module that employs a learnable frequency-domain filter to selectively enhance high-frequency structural cues typically suppressed in tail classes. Second, a geometric balancing module based on polar dual-prototype supervised contrastive (PDP-SupCon) learning, which enforces angular alignment and radial compactness to achieve a uniformly organized embedding space. Third, a statistical balancing module that performs Gaussian tail modeling (GTM) to synthesize variance-regularized features, enriching tail-class diversity and equalizing gradient contributions during optimization. Comprehensive experiments on multiple long-tailed benchmarks, including NWPU-RESISC45-LT (NWPU-LT), AID-LT, and SIRI-WHU-LT, demonstrate that DECOR+ consistently surpasses state-of-the-art methods, particularly under severe imbalance. These results validate the effectiveness of energy-balanced representation learning and highlight the potential of DECOR+ as a principled solution for robust and equitable RS image interpretation. Source code is available at https://github.com/ChloeeGrace/DECOR_PLUS**

***Index Terms*—Contrastive learning, feature enhancement, frequency-domain filter, image classification, long-tailed distribution.**

I. Introduction

REMOTE sensing (RS) imagery provides indispensable observational evidence for understanding large-scale land-surface dynamics. Its interpretation—particularly scene classification and recognition—is central to numerous high-impact applications, such as land-cover mapping, urban development assessment, environmental monitoring, resource exploration, and military intelligence [1], [2], [3], [4], [5], [6], [7], [8], [9]. As the entry point of most RS analytic pipelines, the accuracy of scene classification directly governs the reliability and robustness of subsequent interpretation and decision support systems.

Unlike natural images, RS imagery inherently exhibits a long-tailed distribution due to the uneven occurrence of land-cover types, acquisition limitations, and the high cost of expert annotation. Consequently, common categories (e.g., vegetation or residential areas) appear frequently, while rare categories (e.g., airstrips or specialized infrastructures) are sparsely represented. This leads to a severe class imbalance, as shown in Fig. 1, where head classes dominate both the data quantity and the optimization process. When training deep networks on such imbalanced data, the empirical risk can be formulated as

$$\mathcal{L}_{total} = \sum_{c=1}^{C} \frac{n_c}{N} \mathbb{E}_{x \in c} \left[\ell \left(f_\theta \left(x \right), c \right) \right] \tag{1}$$

where n_c denotes the number of samples in class c and $N = \sum_{c=1}^{C} n_c$. Here, $f_\theta(x)$ denotes the learned representation or prediction of a sample, and $\ell(\cdot, \cdot)$ is a discrepancy metric defined in the prediction space. Since head classes satisfy $n_c \gg n_{c'}$ for most tail classes, their gradients are weighted more heavily during training, causing representations of head categories to dominate while tail-class features remain weakly expressed. This optimization bias forms the foundation of the severe performance degradation commonly observed in long-tailed RS scene recognition.

Consequently, head classes receive disproportionately large gradient updates, dominating the representation learning process and suppressing the learning of tail categories. This

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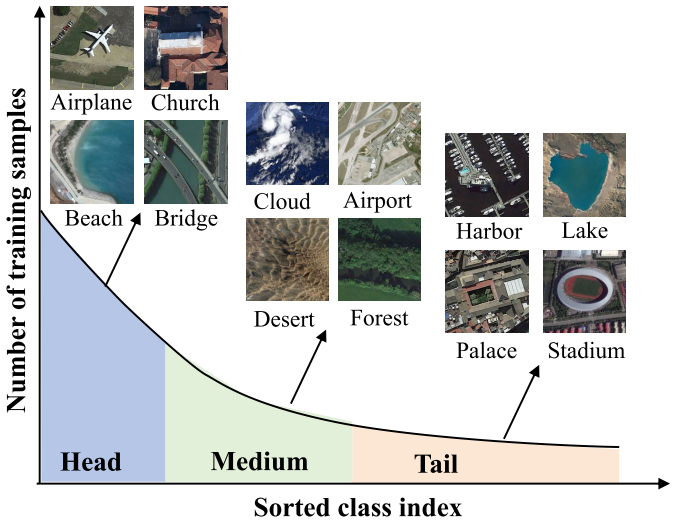

Fig. 1. Long-tailed class distribution. A small number of head classes contain abundant samples, while the majority of tail classes are severely underrepresented, leading to significant data imbalance.

phenomenon can be interpreted as a multidomain energy imbalance: head classes accumulate excessive spectral energy in the frequency domain and form compact, high-magnitude structures in the geometric embedding manifold, while tail classes retain only weak and noisy representations. In addition, gradient dominance biases the classifier decision boundaries in favor of head classes, reinforcing the disparity and leading to persistent degradation in tail recognition. These observations motivate the need for a unified framework capable of restoring spectral, geometric, and gradient energy equilibrium.

To address these challenges, existing studies on long-tailed image classification can be broadly categorized into three directions: representation enhancement, classifier learning, and information enrichment.

1) *Representation Enhancement:* It strengthens the discriminative capacity of feature extractors to produce compact and separable embeddings. Methods, such as affinity loss [10], DLSA [11], and SBCL [12] promote clustering or subclass-aware contrastive learning, while distribution-aware approaches, like DisAlign [13] and SuperDisco [14] explicitly model class discrepancy to reduce embedding overlap. Contrastive paradigms, including KCL [15], TSC [16], PaCo [17], and BCL [18] further regularize geometric separation across classes. However, these methods predominantly operate on the high-level geometric structure of embeddings and often overlook low-level spectral degradation and fairness issues—factors that are particularly critical for RS imagery characterized by rich textures and multiscale patterns.
2) *Classifier Learning:* It focuses on reducing decision bias under skewed distributions. Margin-based techniques such as LDAM [19] and logit-adjustment strategies [20], [21] explicitly reweight classification thresholds, whereas normalization and weight-scaling approaches [22], [23] decouple feature norms from class frequency. Geometry-aware strategies, such as C2AM [24], ECM [25], and curvature regularization [26] further promote fairer classifier manifolds. While effective, these methods mainly target gradient- or decision-level imbalance and do not resolve representation deficiencies arising in the spectral or geometric domains.
3) *Information Enrichment:* It seeks to supplement tail classes through data- or feature-level augmentation. Data mixing strategies, including Mixup, CutMix, M2m, and RSG [27], [28], [29], [30] enhance sample diversity, whereas feature enrichment methods, such as RAC and FASA [31], [32] improve latent representations via retrieval, fusion, or perturbation. Despite improving sparsity, these approaches typically address imbalance locally within the data or feature domain, without providing a unified mechanism to restore equilibrium in spectral energy, geometric representation, and gradient dynamics.

These limitations highlight the need for a holistic framework capable of jointly correcting multidomain energy imbalance—a gap that motivates the design of our proposed DECOR+ framework.

Motivated by these observations, we reinterpret long-tailed RS classification as an imbalance of representation energy across three complementary domains: spectral, geometric, and gradient. Head classes dominate spectral energy by retaining stronger high-frequency responses, forming compact and high-magnitude structures within the geometric embedding manifold, and accumulating disproportionately large gradient energy during optimization. To achieve global equilibrium, the learning objective can be interpreted as jointly addressing imbalance from complementary domains, including spectral representation, embedding geometry, and gradient dynamics

$$\mathcal{L}_{\text{concept}} \triangleq \mathcal{L}_{CE} + \lambda_1 \mathcal{L}_{spec} + \lambda_2 \mathcal{L}_{geom} + \lambda_3 \mathcal{L}_{grad} \tag{2}$$

where $\mathcal{L}_{spec}$, $\mathcal{L}_{geom}$, and $\mathcal{L}_{grad}$ denote the conceptual objectives corresponding to spectral, geometric, and gradient-level balancing, respectively. These objectives are instantiated through architectural design and algorithmic strategies in the proposed framework, rather than implemented as independent loss terms.

Building on this principle, we propose **DECOR+ (energy-balanced representation enhancement)**, a unified framework tailored for long-tailed RS image classification (RSIC). First, a lightweight learnable global filter (LGF) performs spectral balancing by reweighting FFT/IFFT frequency components, amplifying suppressed high-frequency cues of tail classes while preserving global context. Second, a dual-prototype supervised contrastive module restructures the embedding space, enforcing angular consistency and radial compactness to achieve geometric balancing and reduce head-class crowding. Third, a Gaussian tail modeling (GTM) module estimates class-conditional radial distributions and generates variance-regularized synthetic features, enriching tail diversity and restoring gradient energy balance during optimization. The synergy of these components effectively reconstructs spectral–geometric–gradient equilibrium, yielding more robust and equitable RS scene recognition. Extensive experiments demonstrate that DECOR+ consistently improves performance under varying imbalance ratios. Specifically, on the NWPU-LT

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and AID-LT datasets with imbalance ratios of 0.01, 0.02, and 0.1, DECOR+ achieves average performance gains of 7.58% and 18.82%, respectively. On the SIRI-WHU-LT dataset, DECOR+ attains accuracies of 68.47%, 74.17%, and 79.17% under imbalance ratios of 0.01, 0.02, and 0.05, respectively.

The main contributions of this work are summarized as follows.

1) A unified energy-balancing perspective is introduced for long-tailed RS classification, revealing that performance degradation arises from coupled spectral, geometric, and gradient energy imbalance.
2) A spectral balancing module based on an LGF is designed to enhance suppressed high-frequency components of tail classes while preserving global contextual stability.
3) A geometric balancing mechanism is proposed through dual-prototype supervised contrastive learning in polar coordinates, enforcing angular consistency and radial compactness for uniformly separable embeddings.
4) A gradient balancing module is developed using GTM, which statistically estimates class-conditional radial distributions and generates variance-regularized synthetic features to equalize gradient contributions across classes.

II. Related Works

A. RS Image Recognition

As an essential component of RS image interpretation, scene classification has witnessed continuous progress over the past decade, primarily driven by advances in deep learning. Early methods based on convolutional neural networks (CNNs) laid a strong foundation: for example, multiple CNN-based descriptors were fused to improve discriminability in VHR RS scene classification [33]. Multilayer stacked covariance pooling (MSCP) was proposed to aggregate hierarchical convolutional features and enhance representation robustness [34]. A discriminative metric-learning strategy was introduced to jointly optimize CNN features and class separation, significantly improving intraclass compactness and interclass separability [35]. An ensemble framework leveraging a shared backbone with multiple specialized heads was developed to enhance generalization on large-scale benchmarks [36]. More recently, a multiscale CNN was designed to explicitly address scale variation challenges in RS scenes, further strengthening the capabilities of CNN-based models [37].

With the rapid rise of Transformer architectures in computer vision, Transformer-based models have also gained significant traction in RS. A notable example is the multimodal fusion Transformer, which integrates multimodal inputs through cross-patch attention to improve classification accuracy and generalization on challenging datasets [38]. In addition, hierarchical Transformers with multiscale fusion—for instance, the STMSF model—have been proposed to better accommodate spatial variability and high-resolution details inherent to RS imagery, thereby improving scene classification robustness and versatility [39]. The transition from purely CNN-based models toward Transformer-driven architectures continues to expand methodological diversity and push the performance boundary of RS scene understanding.

B. Long-Tailed Distribution

Long-tailed distributions lead to substantial training bias in deep visual recognition, where head classes dominate gradient updates and tail classes receive insufficient supervision. To address this issue, existing studies explore rebalancing strategies from both data and optimization perspectives. Classical data-level solutions include class-balanced loss, which reweights samples based on the effective number of instances to correct gradient imbalance [40]. Mixup [27] and CutMix [28] further enrich minority-class patterns via interpolation or region-level composition, while major-to-minor translation (M2m) [29] explicitly synthesizes tail-like samples to mitigate distribution sparsity. These approaches aim to enhance sample diversity and reduce the dominance of head classes during learning.

Beyond data augmentation, more recent studies focus on feature representation and classifier calibration. Decoupled training (Decouple) [22] proposes separating representation learning from classifier optimization to alleviate the conflict introduced by imbalanced data. Balanced branch networks (BBN) [41] employ dual branch learning to dynamically balance head/tail gradients, and OLTR [42] incorporates a memory-guided feature bank to densify tail-class embeddings. From the classifier perspective, LDAM [19] introduces class-dependent margins to enlarge tail-class separability, while Balanced Softmax [21] and Logit Adjust [20] recalibrate logits with class priors, effectively narrowing the gap between training and test distributions. Although these methods provide strong baselines, long-tailed recognition in RS remains challenging due to complex backgrounds, large intraclass variation, and extreme imbalance across land-cover categories.

C. Contrastive Learning

Contrastive learning has become a powerful framework for representation learning, where the objective is to maximize the agreement between different augmented views of the same instance while separating negatives in the embedding space. Representative methods, such as SimCLR [43], MoCo [44], and SimSiam [45] demonstrate that high-quality representations can be learned through augmentation-driven contrastive objectives. SimCLR relies on large-batch InfoNCE optimization, MoCo introduces a momentum-updated encoder with a dynamic queue for scalable negative sampling, and SimSiam removes the need for negatives entirely through a stop-gradient mechanism that prevents representational collapse.

With the increasing interest in handling imbalance and biased representation learning, contrastive learning has been extended to long-tailed scenarios. Parametric contrastive (PaCo) learning [17] introduces learnable classwise prototypes into the contrastive loss, allowing rare classes to obtain stronger semantic anchors and alleviating the representation sparsity caused by long-tailed distributions. In parallel, self-damaging contrastive learning (SDCL) [46] proposes an intentional feature "damaging" mechanism that suppresses

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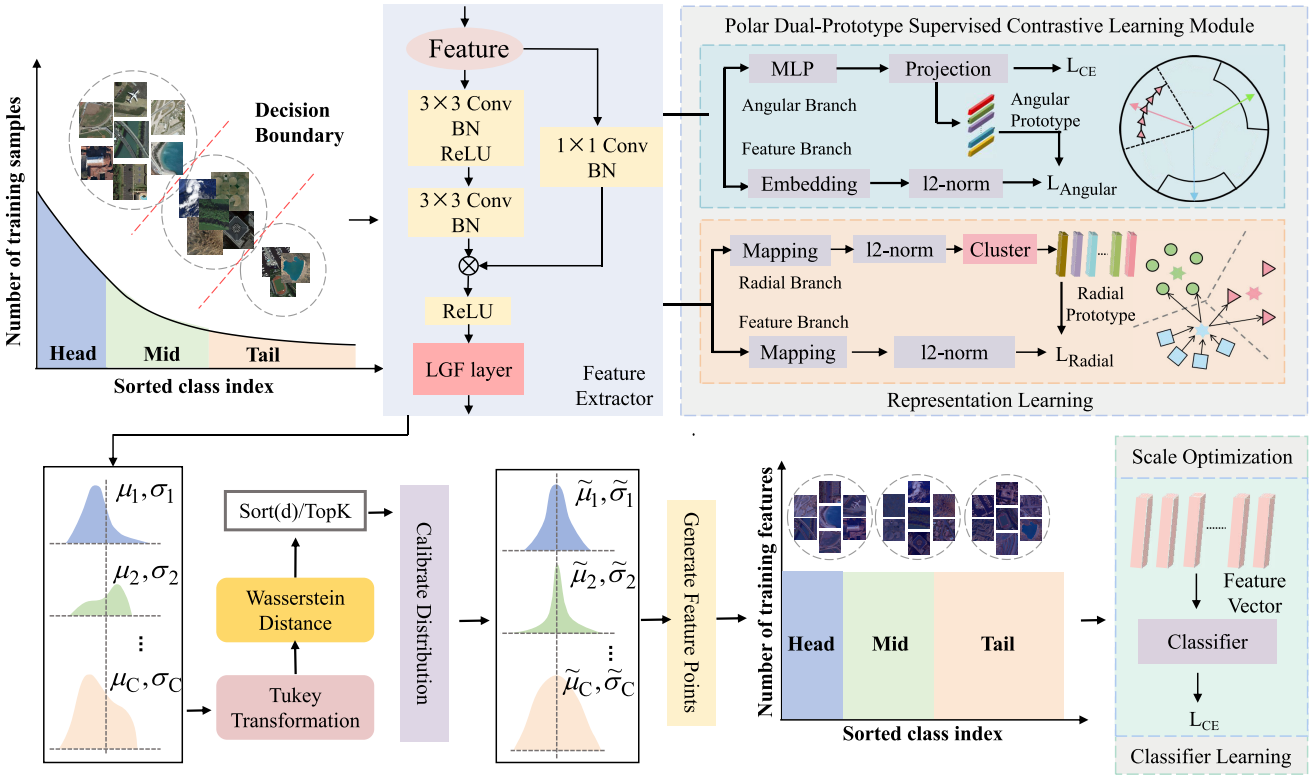

Fig. 2. Overall architecture of DECOR+. DECOR+ balances long-tailed representation learning across spectral, geometric, and gradient domains through LGF, PDP-SupCon, and GTM, jointly improving feature quality, embedding structure, and optimization fairness.

over-confident embeddings, preventing tail-class features from collapsing toward dominant head-class clusters. These methods demonstrate that incorporating class-aware contrastive mechanisms and perturbation-based regularization can substantially improve representation fairness and separability under highly imbalanced data conditions.

III. Methodology

A. Framework Overview and Energy-Balancing Principle

The proposed **DECOR+** framework, as shown in Fig. 2, is developed from a unified energy-balancing perspective, aiming to restore equilibrium in long-tailed representation learning across three complementary domains: spectral energy, geometric energy, and gradient energy. Here, the term energy does not denote a physical quantity, but serves as a unified proxy for classwise representation strength and optimization influence throughout long-tailed learning. Specifically, this notion is instantiated in our framework as spectral energy in feature formation, geometric energy in embedding organization, and gradient energy in optimization dynamics. Under long-tailed distributions, these forms of representation energy become progressively distorted, causing minority classes to be suppressed in early feature extraction, crowded in embedding space, and marginalized during optimization.

To address this issue, DECOR+ adopts a progressive three-stage end-to-end design, where each stage explicitly compensates for one source of energy imbalance while preparing a stable foundation for subsequent stages.

1) *Stage 1: Spectral Energy Balancing:* An LGF is applied to deep network features to rebalance spectral energy during early propagation. By adaptively amplifying discriminative high-frequency components associated with tail classes while preserving low-frequency contextual structure, LGF prevents the premature suppression of minority-class cues and improves the overall quality of feature representations.
2) *Stage 2: Geometric Energy Balancing:* On top of spectrally enhanced features, a polar dual-prototype supervised contrastive (PDP-SupCon) learning module restructures the embedding space from a geometric perspective. By decoupling embedding direction and magnitude through angular and radial constraints, this stage mitigates head-class overconcentration and enforces uniform separability across classes, thereby restoring geometric energy balance in the embedding manifold.
3) *Stage 3: Gradient Energy Balancing:* Finally, a GTM strategy addresses statistical imbalance during optimization. By estimating class-conditional embedding distributions and generating variance-regularized synthetic features for tail classes, GTM increases their effective gradient contributions and alleviates gradient dominance of head classes, leading to fairer and smoother decision boundaries.

Collectively, these three stages form a coherent energy-balancing system that improves representation *quality*, stabilizes feature *distribution*, and promotes decision *fairness*. Through joint optimization, DECOR+ ensures that spectral details, geometric structure, and gradient dynamics are harmonized throughout training, enabling robust long-tailed RSIC.

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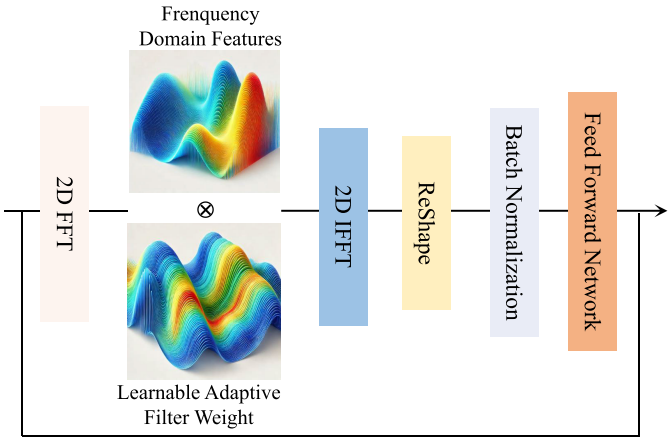

Fig. 3. LGF module. The module adaptively rebalances spectral energy in deep feature maps to enhance feature representation.

B. LGF for Spectral Energy Balancing

In long-tailed learning, tail-class representations are particularly vulnerable during early feature extraction, as limited samples receive insufficient gradient updates and their discriminative cues are easily overwhelmed by dominant head classes. This phenomenon is especially pronounced in RS imagery, where many category-specific structures, such as building edges, linear infrastructures, and small-scale objects, are encoded in high-frequency spectral components. Under long-tailed optimization, head classes tend to absorb a disproportionate amount of spectral energy in early layers, leading to premature suppression of tail-class frequency responses.

To address this issue, we introduce an *LGF* as a spectral energy balancing mechanism that operates on deep feature maps. As shown in Fig. 3, by explicitly reweighting frequency components in a learnable manner, LGF compensates for spectral energy imbalance and enhances discriminative high-frequency cues associated with tail classes before geometric organization takes place.

Given an intermediate feature map $X \in \mathbb{R}^{H\times W\times C}$ extracted from the backbone, an orthonormal 2-D Fourier transform is applied

$$F = \mathrm{FFT2}\,(X) \tag{3}$$

where F encodes both low-frequency structural information and high-frequency details. An LGF $\mathcal{G}$ is then employed to modulate the spectral responses

$$\hat{F} = \mathcal{G} \odot F \tag{4}$$

with $\odot$ denoting elementwise complex multiplication. Unlike fixed or heuristic frequency filters, $\mathcal{G}$ is optimized jointly with the network to adaptively redistribute spectral energy. To maintain stable optimization, $\mathcal{G}$ is initialized close to an identity mapping, so that spectral modulation is introduced progressively rather than abruptly. This allows the model to gradually amplify sparse but discriminative high-frequency components correlated with tail classes while preserving low-frequency contextual information.

The filtered features are reconstructed via inverse orthonormal Fourier transform and normalization

$$\hat{X} = \mathrm{IFFT2}\left(\hat{F}\right), \quad X^{\mathrm{norm}} = \mathrm{Norm}\left(\hat{X}\right) \tag{5}$$

and further refined using a feedforward network with residual connections

$$X^{\mathrm{out}} = \mathrm{FFN}\left(X^{\mathrm{norm}}\right) + X. \tag{6}$$

From an energy-balancing perspective, LGF acts as a dynamic spectral equalizer that redistributes representation energy across frequency bands and classes. Built upon an energy-preserving orthonormal Fourier representation and learned progressively from a near-identity initialization, LGF enables stable spectral reweighting during training. By strengthening the discriminative spectral responses of tail classes at early stages, LGF improves the *quality* of deep features and establishes a balanced spectral foundation for subsequent geometric and gradient-level balancing in the DECOR+ framework.

C. Polar Dual-Prototype Supervised Contrastive Learning

While spectral balancing improves the quality of early features, long-tailed recognition further requires a structured organization of high-level embeddings to prevent geometric collapse of tail classes. In embedding space, long-tailed optimization not only distorts class directions but also induces severe imbalance in representation magnitude, where head classes occupy high-energy regions while tail classes collapse toward low-energy zones. Such an imbalance cannot be fully addressed by conventional supervised contrastive learning, which primarily constrains angular relationships among normalized features.

To explicitly regulate geometric representation energy, we decompose each embedding into angular and radial components. Given an image x_i with label y_i, its embedding $z_i \in \mathbb{R}^D$ is expressed in polar form as

$$u_i = \frac{z_i}{\|z_i\|_2}, \qquad r_i = \|z_i\|_2 \tag{7}$$

where u_i captures directional information, and r_i represents embedding magnitude, which directly reflects representation energy accumulated during optimization.

For each class c, we introduce a *dual prototype* consisting of an angular prototype v_c ($\|v_c\|_2 = 1$) and a radial prototype ρ_c, corresponding to the dominant direction and characteristic energy level of that class. Both prototypes are updated using exponential moving averages of batch statistics, enabling stable and adaptive estimation under imbalanced data.

The angular contrastive loss enforces directional alignment between samples and their class prototype

$$\ell_i^{\mathrm{ang}} = -\log \frac{\exp\left(u_i^\top v_{y_i}/\tau_\theta\right)}{\sum_{c=1}^{C}\exp\left(u_i^\top v_c/\tau_\theta\right)} \tag{8}$$

which promotes intraclass compactness and interclass separation in angular geometry. Complementarily, the radial contrastive loss constrains representation energy by aligning sample magnitudes with classwise radial prototypes

$$\ell_i^{\mathrm{rad}} = -\log \frac{\exp\left(-|r_i-\rho_{y_i}|/\tau_\rho\right)}{\sum_{c=1}^{C}\exp\left(-|r_i-\rho_c|/\tau_\rho\right)} \tag{9}$$

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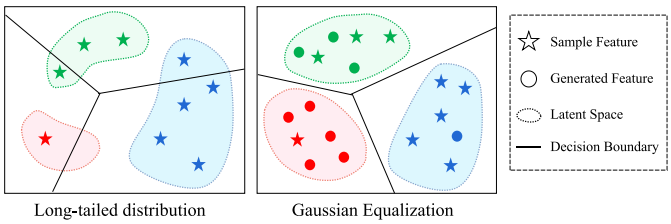

Fig. 4. Representation space and decision boundary under original long-tailed distribution and Gaussian equalization.

thereby preventing head classes from monopolizing high-energy regions while encouraging tail classes to acquire sufficient representation strength.

The overall geometric balancing objective is formulated as

$$\mathcal{L}_{\text{PDP-SupCon}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \left(\ell_i^{\text{ang}} + \lambda \, \ell_i^{\text{rad}} \right) \tag{10}$$

where $\mathcal{B}$ denotes the training batch, and λ controls the contribution of radial energy regulation.

From a geometric energy perspective, PDP-SupCon decouples directional discrimination from magnitude allocation, enabling representation energy to be redistributed more evenly across classes. By jointly regulating angular structure and radial energy, this module alleviates head-class overconcentration in the embedding manifold and establishes a balanced geometric foundation for subsequent gradient-level correction in the DECOR+ framework.

D. GTM for Gradient Energy Balancing

Although geometric energy balancing improves the organization of embedding space, long-tailed optimization still suffers from severe imbalance at the statistical level. Head classes, with abundant samples, dominate gradient accumulation and induce large-margin decision regions, while tail classes contribute sparse and high-variance gradients that are insufficient to shape stable decision boundaries. This discrepancy reveals that geometric alignment alone cannot fully restore equilibrium in gradient energy distribution.

To address this issue, we introduce *GTM*, a statistical mechanism designed to rebalance classwise gradient energy by compensating for density sparsity in tail classes. Rather than directly augmenting data, GTM estimates class-conditional embedding distributions and generates variance-regularized synthetic features to stabilize gradient contributions during classifier optimization.

A classwise Gaussian distribution is constructed for each category. For class C_k, the feature mean and covariance matrix are computed as

$$\mu_k = \frac{1}{N_k} \sum_{z_i \in D_k} z_i, \quad \Sigma_k = \frac{1}{N_k} \sum_{z_i \in D_k} (z_i - \mu_k)(z_i - \mu_k)^\top \tag{11}$$

where N_k denotes the number of samples in class C_k, and D_k represents the feature set of class C_k.

To encourage the feature distribution to better approximate a Gaussian form, each feature vector z_i is transformed using Tukey's Ladder of Powers

$$\tilde{z}_i = \begin{cases} z_i^\lambda, & \lambda \neq 0 \\ \log(z_i), & \lambda = 0 \end{cases} \tag{12}$$

where λ is a hyperparameter, typically set to 1. This transformation alleviates skewness and promotes distributional normality.

For each sample, the Mahalanobis distance to all class distributions is computed as

$$d_M(\tilde{z}_i, C_k) = (\tilde{z}_i - \mu_k)^\top \Sigma_k^{-1} (\tilde{z}_i - \mu_k) \tag{13}$$

and the K nearest categories are selected to form the neighborhood set

$$S_i = \{k \mid k \in \{1, 2, \ldots, L\} \; k \neq c_i, \, d_M(\tilde{z}_i, C_k) \in \text{top-}K\,[d_M]\} \tag{14}$$

where c_i denotes the ground-truth class of sample z_i, and L is the total number of categories. This neighborhood set is used to assist the calibration of tail classes, preventing tail samples from forming isolated distributions.

Based on the neighboring category set S_i, distribution calibration is performed by computing the weighted mean and covariance with weight normalization

$$\mu_{\tilde{z}_i} = \sum_{k \in S_i} w_k \mu_k, \quad \Sigma_{\tilde{z}_i} = \sum_{k \in S_i} w_k \Sigma_k + \alpha I \tag{15}$$

where the weighting coefficient w_k is defined using a softmax over the negative Mahalanobis distances

$$w_k = \frac{\exp(-d_k)}{\sum_{j \in \text{top-}K} \exp(-d_j)}, \quad d_k = d_M(\tilde{z}_i, C_k) \tag{16}$$

and α is a small diagonal regularization coefficient used to stabilize the calibrated covariance. It ensures positive definiteness, prevents variance collapse along under-observed directions, and imposes a mild isotropic uncertainty floor, thereby avoiding overconfident distribution estimation under extreme sample scarcity.

Let the original feature set be denoted as

$$\mathcal{F} = \{z_i\}_{i=1}^N. \tag{17}$$

New features are sampled from the calibrated Gaussian distribution

$$z_{\text{gen}} \sim \mathcal{N}\left(\mu_{\tilde{z}_i}, \Sigma_{\tilde{z}_i}\right) \tag{18}$$

and merged with the original feature set

$$\mathcal{F} = \mathcal{F} \cup \{z_{\text{gen}}\}. \tag{19}$$

The balanced feature set is finally fed into the classifier for training. This procedure increases the feature density and diversity of tail classes, alleviates the dominance of head classes in the decision space, and leads to a more equitable and robust classification boundary, as shown in Fig. 4.

By increasing the effective density of tail classes in embedding space, GTM enhances their gradient contribution without altering the original data distribution.

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Algorithm 1 Progressive Representation Energy Balancing for Long-Tailed RS Classification

Input: Training data $\mathcal{D} = \{(x_i, y_i)\}$; batch size B; number of classes C.
Input: Learnable parameters θ (backbone), ψ (projection head), $\mathcal{G}$ (LGF).
Input: Class prototypes $\{v_c, \rho_c\}_{c=1}^{C}$ initialized by batch statistics.
for $t = 1$ **to** `max_iter` **do**
  Sample a mini-batch $\mathcal{B} = \{(x_i, y_i)\}_{i=1}^{B}$ from $\mathcal{D}$;
  `// Stage 1: Spectral energy balancing via LGF`
  Extract deep feature maps $X = f_\theta(x_i)$;
  Apply FFT2, learnable filter $\mathcal{G}$, and IFFT2 to obtain spectrally enhanced features X^{out};
  `// Stage 2: Geometric energy balancing via PDP-SupCon`
  Compute embeddings $z_i = g_\psi(X^{out})$;
  Decompose embeddings into angular and radial components (u_i, r_i);
  Update angular and radial prototypes $\{v_c, \rho_c\}$ using exponential moving averages;
  Compute $\mathcal{L}_{\text{PDP-SupCon}}$ using angular and radial contrastive objectives;
  `// Stage 3: Statistical gradient balancing via GTM`
  Estimate classwise statistics (μ_c, Σ_c) from current embeddings;
  Perform local distribution calibration for tail classes and sample synthetic features $\tilde{z}$;
  `// Classifier optimization`
  Compute classification loss $\mathcal{L}_{cls}$ using both real and synthetic features;
  Compute total loss $\mathcal{L}_{total} = \mathcal{L}_{cls} + \beta\mathcal{L}_{\text{PDP-SupCon}}$;
  Update learnable parameters $\{\theta, \psi, \mathcal{G}\}$ via backpropagation;
end

From an optimization perspective, this process reduces classwise gradient variance and drives the expected gradient norm toward equilibrium

$$\mathbb{E}_c\left[\|\nabla_\theta \mathcal{L}_c\|\right] \approx \text{constant across } c \tag{20}$$

thereby preventing head classes from monopolizing optimization dynamics. In RS scenarios, where rare land-cover categories often exhibit large intraclass variability, GTM stabilizes learning and yields smoother, fairer decision boundaries under long-tailed conditions.

E. Energy Interpretation and Optimization Perspective

From an optimization standpoint, long-tailed learning can be interpreted as a process of imbalanced representation strength accumulation across classes. To provide a unified explanation for the proposed framework, we introduce an *energy-based interpretation* that characterizes how representation dominance emerges and propagates under long-tailed optimization.

Let the representation energy of class c be defined as

$$E_c = \mathbb{E}_{x\sim\mathcal{D}_c}\left[\|f_\theta(x)\|_2^2\right] \tag{21}$$

where $f_\theta(\cdot)$ denotes the feature extractor, and $\mathcal{D}_c$ represents the data distribution of class c. This quantity reflects the average embedding magnitude of class c and serves as a proxy for its representation strength and accumulated optimization influence.

Under long-tailed distributions, head classes possess substantially larger sample counts and contribute more frequently to gradient updates, resulting in significantly higher representation energy

$$E_{\text{head}} \gg E_{\text{tail}}. \tag{22}$$

This disparity manifests across multiple stages of learning, including early spectral responses, geometric organization of embeddings, and classwise gradient accumulation, ultimately leading to biased decision boundaries.

Rather than explicitly minimizing a physical energy function, the proposed framework aims to *redistribute representation strength* across complementary domains through coordinated loss design. Specifically, DECOR+ performs energy compensation at three levels as follows.

1) Spectral representation strength is enhanced by the LGF, which rebalances frequency responses and prevents early suppression of tail-class features.
2) Geometric representation strength is regulated by PDP-SupCon learning, which decouples directional discrimination from magnitude allocation and alleviates classwise overconcentration in embedding space.
3) Statistical optimization strength is compensated by GTM, which increases effective tail-class density and stabilizes gradient contributions during classifier learning.

These components collectively guide the optimization process toward a balanced regime in which no single class dominates representation learning. For interpretability, we summarize the overall representation strength using a three-part decomposition

$$E_{\text{total}} \triangleq E_{\text{spec}} + E_{\text{geom}} + E_{\text{stat}} \tag{23}$$

where E_{spec}, E_{geom}, and E_{stat} denote the spectral, geometric, and statistical (optimization) components of representation strength, respectively. Note that this decomposition serves as an explanatory proxy rather than a physically minimized energy function; the balance is encouraged implicitly through the coordinated design of LGF, PDP-SupCon, and GTM. During training, minimizing the unified objective function (see Algorithm 1) implicitly encourages these components to evolve toward a more balanced configuration, resulting in improved recognition performance for tail classes without sacrificing head-class accuracy.

IV. Experiments

A. Datasets

To evaluate the effectiveness of the proposed DECOR+ framework under realistic long-tailed conditions, we reorganize several widely used RSIC benchmarks into imbalanced

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TABLE I
DETAILED INFORMATION ABOUT NWPU-LT

Classname	Training Samples			Classname	Training Samples		
	imb=0.01	*imb*=0.02	*imb*=0.1		*imb*=0.01	*imb*=0.02	*imb*=0.1
Overpass	600	600	600	Intersection	540	548	569
Rectangular Farmland	486	502	540	Desert	438	459	512
Wetland	394	420	486	Industrial Area	355	384	461
Circular Farmland	320	351	438	Terrace	288	322	415
Harbor	259	294	394	Medium Residential	233	269	374
Railway	210	246	355	Church	189	225	337
Palace	170	206	320	Roundabout	153	188	303
Baseball diamond	138	172	288	Cloud	124	158	273
Tennis Court	112	144	259	Chaparral	101	132	246
Parking Lot	91	121	233	Island	82	110	221
Runway	73	101	210	Airport	66	92	199
Stadium	60	84	189	Freeway	54	77	180
Forest	48	71	170	Golf Course	43	64	162
Meadow	39	59	153	Basketball Court	35	54	146
Beach	32	49	138	Ground Track Field	28	45	131
Lake	25	41	124	Commercial Area	23	38	118
Mobile Home Park	21	34	112	Ship	18	31	106
Railway Station	17	29	101	Snowberg	15	26	96
Dense Residential	13	24	91	Sea Ice	12	22	86
Airplane	11	20	82	Bridge	10	18	77
Storage Tank	9	17	73	Mountain	8	15	70
Sparse Residential	7	14	66	Thermal Power Station	6	13	63
River	6	12	60	Total	5962	6901	10627

TABLE II
DETAILED INFORMATION ABOUT AID-LT

Classname	Training Samples			Classname	Training Samples		
	imb=0.01	*imb*=0.02	*imb*=0.1		*imb*=0.01	*imb*=0.02	*imb*=0.1
Airport	100	100	100	Bareland	85	87	92
Baseball Field	72	76	85	Beach	62	66	78
Bridge	52	58	72	Center Area	45	50	67
Church	38	44	62	Commercial	32	38	57
Dense Residential	28	33	52	Desert	23	29	48
Farmland	20	25	45	Forest	17	22	41
Industrial	14	19	38	Meadow	12	17	35
Medium Residential	10	15	32	Mountain	9	13	30
Park	7	11	28	Parking Lot	6	10	25
Playground	5	8	23	Pound	4	7	22
Port	3	6	20	Railway Station	3	5	18
Resort	2	5	17	River	2	4	16
School	1	3	14	Sparse Residential	1	3	13
Square	1	2	12	Stadium	1	2	11
Storage Tanks	1	2	10	Viaduct	1	2	10

TABLE III
DETAILED INFORMATION ABOUT SIRI-WHU-LT

Classname	Training Samples			Classname	Training Samples		
	imb=0.01	*imb*=0.02	*imb*=0.05		*imb*=0.01	*imb*=0.02	*imb*=0.05
Agriculture	140	140	140	Overpass	92	98	106
Residential	60	68	81	Commercial	39	48	61
Pond	26	33	47	Idle Land	17	23	35
Harbor	11	16	27	Industrial	7	11	20
Meadow	4	8	15	Water	3	5	12
Park	2	3	9	River	1	2	7

training sets. Rather than artificially enforcing uniform class distributions, we emulate the inherent long-tailed property commonly observed in RS scenarios, where a small number of land-cover categories dominate the data while many others appear sparsely.

Specifically, categories are sorted in descending order according to their sample sizes for evaluation, as summarized in Tables I–III. This protocol enables a fine-grained assessment of model performance across classes with varying data availability.

Following standard practice in long-tailed learning, the number of training samples for the ith class is defined as follows:

$$N_i = N_{\max} \cdot \text{imb}^{\frac{i-1}{C-1}} \tag{24}$$

where $N_{\max}$ denotes the maximum class size, C is the total number of categories, and imb controls the imbalance ratio between the largest and smallest classes. According to the class frequency distribution, categories with sample numbers exceeding $0.75N_{\max}$ are defined as *head classes*, while those with fewer than $0.15N_{\max}$ are regarded as *tail classes*. The remaining categories are treated as *medium-frequency classes*. This exponential allocation ensures that head classes dominate the training data while tail classes contain only a limited number of samples. We consider four imbalance ratios, $\text{imb} \in \{0.01, 0.02, 0.05, 0.1\}$, corresponding to increasingly severe long-tailed distributions.

Through this construction, the resulting datasets reflect realistic RS scenarios in which certain semantic categories are abundantly represented, whereas many others appear infrequently. This provides a faithful testbed for evaluating the robustness of long-tailed recognition methods.

1) NWPU-LT: The NWPU-RESISC45 dataset [47] is one of the most widely used benchmarks for RS scene classification, consisting of 45 semantic categories with 700 samples each, resulting in a total of 31 500 images of 256×256 pixels. Following the conventional split, 100 samples per class are held out for testing, and the remaining 600 images are adopted for long-tailed sampling. In (24), $C = 45$ denotes the total number of categories, $N_{\max} = 600$ represents the maximum class size, and $\text{imb} \in \{0.01, 0.02, 0.1\}$ is the imbalance coefficient that controls the ratio between head and tail categories. This procedure produces the NWPU-LT variants, which are summarized in Table I.

2) AID-LT: The AID dataset [48] is another widely used benchmark for aerial scene classification, comprising 30 semantic categories with image resolutions of 600×600 pixels. The number of samples per class varies between 220 and 420. Following the same evaluation protocol, 100 samples per category are held out for testing, while the remaining images are reorganized to form long-tailed distributions using the exponential allocation strategy. In (24), $C = 30$ denotes the total number of categories, $N_{\max} = 100$ represents the maximum class size, and $\text{imb} \in \{0.01, 0.02, 0.1\}$ is the imbalance coefficient that controls the ratio between head and tail categories. This procedure produces the AID-LT variants, which are summarized in Table II.

3) SIRI-WHU-LT: The SIRI-WHU dataset [49] is a widely used benchmark for RS scene classification, containing 12 semantic categories with 200 images per class, yielding a total of 2400 high-resolution images of 200×200 pixels. Following the conventional protocol, 60 samples per category are held out for testing, and the remaining 140 images are utilized for constructing long-tailed splits. In (24), $C = 12$ denotes the total number of categories, $N_{\max} = 140$ represents the maximum class size, and $\text{imb} \in \{0.01, 0.02, 0.05\}$ is the imbalance coefficient that controls the ratio between head and tail categories. This procedure produces the SIRI-WHU-LT variants, which are summarized in Table III.

B. Implementation Details and Evaluation Metrics

1) Implementation Details: All experiments are implemented using PyTorch and conducted on four NVIDIA A6000 GPUs. Unless otherwise specified, the proposed

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TABLE IV
TOP-1 ACCURACY (%) ON NWPU-LT

NWPU-LT	Backbone	Imbalance ratio											
		100:1				50:1				10:1			
		Head	Middle	Tail	ALL	Head	Middle	Tail	ALL	Head	Middle	Tail	ALL
Baseline	GoogleNet [52]	97.00	92.00	64.42	76.40	97.75	91.67	70.04	81.16	96.17	90.87	87.38	90.29
	MobileNet-V2 [53]	98.67	93.00	68.08	78.98	98.00	94.83	75.87	85.42	96.50	91.19	84.88	90.78
	ResNet-18 [50]	96.67	92.75	64.58	76.73	98.75	93.17	71.74	82.71	95.50	89.74	82.63	89.24
	ResNet-50 [50]	98.00	94.56	68.77	79.88	98.50	94.72	76.74	85.84	97.00	92.00	84.50	91.33
LDAM [19]	Resnet-50 [50]	98.33	93.50	72.50	81.69	**99.00**	94.06	79.22	86.91	94.33	92.52	87.00	91.78
BKD [54]		98.00	94.88	69.31	80.31	96.50	93.61	81.52	87.69	96.17	91.65	89.88	91.93
CB [40]		95.67	90.75	73.88	81.33	96.00	92.83	79.26	86.18	97.00	92.32	87.13	92.02
LogitAdjustment [20]		**99.33**	95.00	71.65	81.80	98.25	95.22	77.52	86.44	97.17	92.77	88.25	92.56
RSG [30]		98.00	92.69	73.42	81.91	98.50	94.83	78.26	86.69	97.50	92.23	88.88	92.33
Hybrid [55]		97.33	95.44	73.58	82.93	97.75	94.61	80.84	87.67	96.17	94.03	88.63	93.36
MiSLAS [56]		97.00	93.38	81.16	86.56	98.25	94.17	85.44	90.07	**97.67**	94.39	93.38	94.64
LTRL [57]		92.00	89.44	75.35	82.46	93.50	90.78	82.61	86.84	94.50	92.77	91.75	92.82
ProCo [58]		96.67	92.63	80.31	85.78	97.25	94.28	84.39	89.49	96.67	93.29	91.00	93.33
LIFT [59]	ViT-B/16 [60]	94.67	91.69	88.08	89.80	95.00	92.28	**91.65**	92.20	94.00	92.32	94.25	92.89
DECOR [61]	GoogleNet [52]	97.33	93.75	76.54	84.04	97.00	94.39	82.04	88.31	96.83	93.77	89.50	93.42
	MobileNet-V2 [53]	96.33	95.25	79.54	86.24	97.25	95.00	84.96	90.07	96.50	94.06	89.63	93.60
	ResNet-18 [50]	94.67	93.75	77.92	84.67	97.50	94.00	82.83	88.60	96.83	93.13	90.13	93.09
	ResNet-50 [50]	97.33	**95.19**	85.38	89.67	97.25	**95.94**	88.39	92.20	97.50	**95.26**	92.13	95.00
DECOR+(ours)	GoogleNet [52]	97.67	91.88	80.08	85.17	97.00	92.33	84.96	88.98	96.33	94.16	92.50	94.15
	MobileNet-V2 [53]	95.33	94.25	82.27	87.40	97.25	95.28	85.35	90.38	96.33	94.266	91.13	93.97
	ResNet-18 [50]	94.00	92.75	80.42	85.71	96.00	93.83	84.30	89.15	95.83	93.54	91.00	93.40
	ResNet-50 [50]	95.33	92.56	**90.54**	**91.58**	98.00	93.83	91.52	**93.02**	96.00	95.10	**95.00**	**95.20**

DECOR+ framework is built upon a ResNet-50 backbone [50] to ensure fair comparison with existing long-tailed recognition methods. Input images are resized to 224×224 pixels.

The network is trained for 100 epochs using stochastic gradient descent (SGD) [51] with a momentum of 0.9 and a weight decay of 1×10^{-4}. The batch size is set to 64, and the initial learning rate is fixed at 0.005. A cosine annealing schedule is employed to adaptively decay the learning rate during training.

For all comparative experiments, identical training protocols and optimization settings are adopted across different methods unless explicitly stated, ensuring a fair and consistent evaluation under long-tailed conditions.

C. Evaluation Metrics

To comprehensively assess the effectiveness and efficiency of the proposed method, we adopt a set of quantitative and qualitative evaluation metrics that reflect both recognition performance and representational behavior under long-tailed distributions.

1) Top-1 Accuracy: Top-1 accuracy is employed as the primary quantitative metric to evaluate classification performance. It measures the proportion of test samples whose predicted label with the highest confidence matches the ground truth, reflecting the overall recognition capability of the model across all categories.

2) T-SNE Visualization: To analyze the geometric structure of learned representations, we visualize high-dimensional feature embeddings using t-distributed stochastic neighbor embedding (t-SNE). This visualization illustrates intraclass compactness and interclass separability, providing insight into how long-tailed imbalance is alleviated in the embedding space.

3) FLOPs per Second: Floating-point operations (FLOPs) are adopted to quantify the computational cost of a model. Specifically, FLOPs reflect the number of floating-point operations required during inference and are widely used to evaluate the algorithmic complexity and efficiency of deep learning models.

4) Confusion Matrix: Confusion matrices are used to examine classwise prediction behavior, particularly for tail categories. Each row corresponds to ground-truth labels and each column to predicted labels, allowing a detailed analysis of misclassification patterns under imbalanced conditions.

5) Attention Map Visualization: To further investigate the spatial perception ability of the model, we visualize attention maps using Grad-CAM [68]. These maps highlight image regions that contribute most significantly to classification decisions, offering qualitative evidence of how DECOR+ enhances discriminative focus, especially for small or structurally complex RS targets.

D. Comparison Results

We further benchmark DECOR+ against a range of representative long-tailed recognition approaches, including LDAM [19], BKD [54], CB [40], Logit Adjustment [20], RSG [30], Hybrid [55], MiSLAS [56], LRTR [57], ProCo [58], and LIFT [59]. The comparison is conducted under three imbalance levels on both NWPU-LT and AID-LT. As summarized in Table IV, DECOR+ built upon ResNet-50 consistently surpasses the baseline by margins of 11.70%, 7.18%, and 3.87% at imbalance ratios of 0.01, 0.02, and 0.1, respectively. Notably, the relative improvement becomes more pronounced as the degree of imbalance intensifies, suggesting that DECOR+ is particularly advantageous in highly skewed data scenarios. In addition, when deployed on alternative backbones, such as GoogleNet [52], MobileNet-V2 [53], and ResNet-18 [50], our method continues to exhibit substantial gains, confirming its broad applicability across different network architectures.

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TABLE V
TOP-1 ACCURACY (%) ON AID-LT

AID-LT	Backbone	Imbalance ratio											
		100:1				50:1				10:1			
		Head	Middle	Tail	ALL	Head	Middle	Tail	ALL	Head	Middle	Tail	ALL
Baseline	GoogleNet [52]	92.50	93.20	35.17	58.33	95.67	87.82	40.50	63.37	93.25	83.60	69.00	81.96
	MobileNet-V2 [53]	86.00	90.20	43.56	61.93	94.00	89.18	42.13	64.57	83.60	83.70	71.67	82.57
	ResNet-18 [50]	90.50	91.50	35.78	58.00	91.67	87.36	40.13	62.60	90.25	82.50	66.50	80.33
	ResNet-50 [50]	95.00	91.30	38.94	60.13	95.67	90.36	41.13	64.63	94.00	84.85	72.00	83.50
LDAM [19]	Resnet-50 [50]	87.50	88.60	43.06	61.20	95.33	88.19	54.50	71.13	94.50	84.55	74.33	83.83
LogitAdjustment [20]		91.00	93.90	40.00	61.37	94.33	92.64	47.91	68.90	93.00	85.55	75.67	84.57
CB [40]		90.50	92.70	44.94	63.90	94.33	91.73	46.38	67.80	93.25	86.65	75.67	85.33
MiSLAS [56]		91.50	88.60	50.44	65.90	90.33	86.73	62.75	74.30	87.75	85.30	82.67	85.10
BKD [54]		96.00	**96.20**	51.22	69.20	96.33	91.27	55.19	72.53	93.50	87.85	83.33	87.70
RSG [30]		84.50	94.90	53.94	69.63	91.33	89.27	63.63	75.80	93.75	86.20	77.33	85.43
Hybrid [55]		96.50	95.70	52.67	69.93	97.00	94.18	62.56	77.60	97.75	87.80	79.83	87.53
ProCo [58]		89.00	89.30	60.72	72.13	93.67	86.75	71.93	80.03	95.50	86.50	74.83	85.37
LIFT [59]	VIT-B/16 [60]	**98.50**	94.70	49.11	67.60	97.00	94.64	56.63	74.60	**98.25**	91.65	87.50	91.70
DECOR [61]	GoogleNet [52]	91.00	95.20	68.83	79.10	94.00	94.09	74.38	83.56	95.00	90.75	83.83	89.93
	MobileNet-V2 [53]	94.00	94.60	65.88	77.33	96.33	95.45	75.06	84.67	95.75	90.75	83.67	90.00
	ResNet-18 [50]	91.00	94.20	65.94	77.03	94.00	93.18	75.68	83.93	96.00	89.25	85.33	89.36
	ResNet-50 [50]	95.50	95.80	72.16	81.60	**97.33**	**95.55**	79.63	87.23	97.25	91.75	86.83	91.50
DECOR+(ours)	GoogleNet [52]	92.00	93.30	70.44	79.50	94.67	93.36	76.31	84.40	95.75	91.40	85.33	90.77
	MobileNet-V2 [53]	88.50	94.60	68.22	78.37	94.00	94.09	78.63	85.83	96.00	90.85	86.17	90.60
	ResNet-18 [50]	91.50	94.40	67.61	78.13	97.00	93.09	77.25	85.03	95.25	91.60	83.33	90.43
	ResNet-50 [50]	95.00	93.40	**76.56**	**83.40**	95.33	94.19	**83.63**	**88.67**	96.75	**92.40**	**90.83**	**92.67**

For a fair comparison, all competing long-tailed methods are evaluated on ResNet-50. Loss reweighting schemes such as LDAM and CB adjust the decision boundary but do not fundamentally enhance the expressive capacity of the feature extractor, leading to limited improvements. Logit adjustment places greater emphasis on minority classes, and BKD leverages distillation to promote tail learning, though their gains on the tail classes remain modest due to insufficient feature-level refinement. Hybrid introduces a contrastive component that strengthens intermediate representations yet still underperforms on severely imbalanced subsets. RSG mitigates imbalance by augmenting head classes; however, its mechanism offers only partial benefits for underrepresented categories. LTRL focuses on improving tail representations for long-tailed recognition, but its effectiveness remains constrained under highly imbalanced settings. ProCo introduces prototype-based contrastive learning to pull samples toward class prototypes, enhancing intraclass compactness, but its effectiveness depends heavily on reliable prototype estimation when tail data are scarce. LIFT employs feature interpolation to enrich tail-class samples and increase representation diversity; nevertheless, when the number of tail samples is extremely limited, the interpolated features may deviate from the true data manifold. Consequently, although these approaches outperform traditional reweighting-based methods, they remain limited in fully correcting feature collapse and distribution skewness in highly imbalanced scenarios.

In contrast, DECOR+ simultaneously strengthens global feature encoding through a learnable frequency-domain filter and achieves distribution-level refinement via PDP-SupCon learning and GTM. This combination substantially improves the discriminative quality of tail-class embeddings. As shown in Table V, the performance trends observed on NWPU-LT are consistently reproduced on AID-LT. Under the same ResNet-50 backbone, DECOR+ yields improvements of 23.27%, 24.04%, and 9.17% at three imbalance levels. Although MiSLAS exhibits slightly better performance on NWPU-LT in certain settings, it degrades notably on the smaller AID dataset, suggesting limited generalizability. DECOR+ maintains strong performance across datasets and imbalance conditions, demonstrating its robustness and stability relative to existing techniques.

TABLE VI
TOP-1 ACCURACY (%) ON THE LONG-TAILED SIRI-WHU-LT DATASET

Method	Imbalance ratio		
	0.01	0.02	0.05
LDAM [19]	43.77	35.56	59.84
CB [40]	49.98	54.09	67.81
BBN [41]	51.80	54.86	68.75
BKD [54]	46.33	65.72	70.51
BCL [18]	42.73	54.81	68.19
CSA [62]	55.21	62.08	73.33
AREA [63]	54.49	68.75	72.59
CCSMLW [64]	50.71	54.91	67.35
GLMC [65]	45.07	56.43	72.45
Mixup [27]	34.65	47.27	50.18
CutMix [28]	44.81	53.19	69.03
DIFFUSEMIX [66]	60.27	64.23	68.68
DONA [67]	65.42	71.85	77.68
DECOR [61]	66.25	72.50	78.19
DECOR+	**68.47**	**74.17**	**79.17**

We further conduct evaluations on an additional small-scale dataset, and the results are presented in Table VI. Under imbalance levels of 0.01, 0.02, and 0.05, DECOR+ achieves accuracies of 68.47%, 74.17%, and 79.17%, outperforming other methods by margins ranging from 2.22%–33.82%, 1.67%–38.61%, and 0.98%–28.99%, respectively. While methods, such as CSA, AREA, GLMC, and CutMix, deliver

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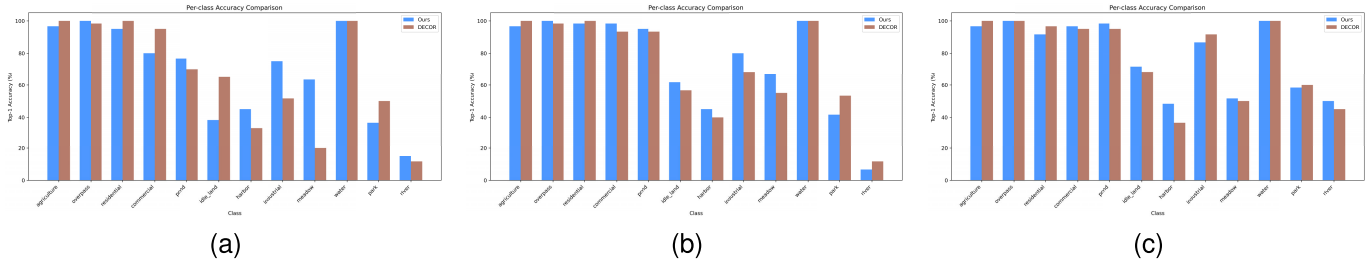

Fig. 5. Comparison of Top-1 accuracy under different imbalance ratios (IR) on the SIRI-WHU-LT dataset. (a) IR = 0.01. (b) IR = 0.02. (c) IR = 0.05.

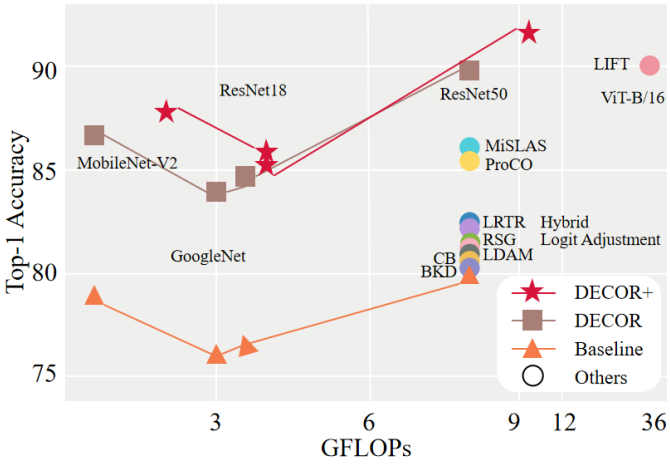

Fig. 6. Complexity/accuracy tradeoffs for baseline and our model on the NWPU-LT dataset with an imbalance ratio of 0.01.

competitive results under mild imbalance, their performance drops sharply as the imbalance ratio grows. Although generative augmentation approaches such as DIFFUSEMIX and DONA can also achieve competitive performance, they suffer from inherent limitations, including the distributional mismatch between synthesized and real images and the limited effectiveness of generating reliable samples for extremely scarce tail classes. Overall, these results confirm that DECOR+ offers a more effective and reliable solution to long-tailed recognition across diverse datasets and imbalance severities. Fig. 5 presents the classwise Top-1 accuracy on the SIRI-WHU-LT dataset under three imbalance factors, i.e., γ = 0.01, 0.02, and 0.05. Compared with DECOR, the proposed method consistently achieves higher recognition accuracy for tail categories across all imbalance settings. Notably, under the most severe imbalance condition (γ = 0.01), where only a very limited number of samples are available for minority classes, our approach maintains stable performance across most categories. This suggests that the proposed method effectively alleviates the dominance of head classes and retains discriminative capability for tail classes, even in extremely skewed data distributions.

To further assess the efficiency of our method, we compare the computational complexity of the baseline model, DECOR+, and several representative long-tailed recognition approaches, as shown in Fig. 6. Although DECOR+ incurs a relatively higher computational cost due to the proposed global enhancement module, its inference process remains concise, introducing only a small number of additional parameters and avoiding auxiliary components such as the PDP-SupCon module. More importantly, the additional computation translates into a substantial performance improvement, demonstrating that the accuracy gain brought by DECOR+ far outweighs its moderate increase in FLOPs. This reflects a highly favorable accuracy–efficiency tradeoff within the inference process.

To further assess the efficiency of our method, we compare the computational complexity of the baseline model, DECOR+, and several representative long-tailed recognition approaches, as shown in Fig. 6. Although DECOR+ incurs a relatively higher computational cost due to the proposed global enhancement module, its inference process remains concise, introducing only a small number of additional parameters and avoiding auxiliary components such as the PDP-SupCon module. More importantly, the additional computation translates into a substantial performance improvement, demonstrating that the accuracy gain brought by DECOR+ far outweighs its moderate increase in FLOPs. This reflects a highly favorable accuracy–efficiency tradeoff within the inference process.

To further elucidate the advantages brought by DECOR+, we perform a comprehensive feature visualization analysis on the NWPU-LT and AID-LT datasets under the severe imbalance setting (γ = 0.01). As illustrated in Figs. 7 and 8, the representations produced by DECOR+ exhibit markedly enhanced intraclass compactness and clearer interclass boundaries compared with those obtained using standard cross-entropy training. This improvement in the geometric structure of the embedding space indicates that DECOR+ facilitates more discriminative feature learning, particularly for minority categories.

In addition to representation visualization, we present confusion matrices for both the baseline ResNet-50 and DECOR+(resnet50) counterpart, as shown in Figs. 9 and 10. The comparison highlights the substantial performance gap in tail categories. For the NWPU-LT dataset, the baseline accurately classifies only 11 samples in the “River” category, whereas DECOR+ identifies 74 samples, demonstrating a significant enhancement in rare-class discrimination. Similarly, on the AID-LT dataset, the baseline model recognizes merely one sample in “Stadium” and four in “Storage Tanks.” In contrast, DECOR+ correctly classifies 65 and 97 samples, respectively. Even for the most challenging categories, such as “Sparse Residential” and “Square,” where the baseline fails entirely, DECOR+ successfully retrieves positive predictions, indicating a clear breakthrough.

Overall, these visual and quantitative analyses consistently demonstrate that DECOR+ not only elevates the global

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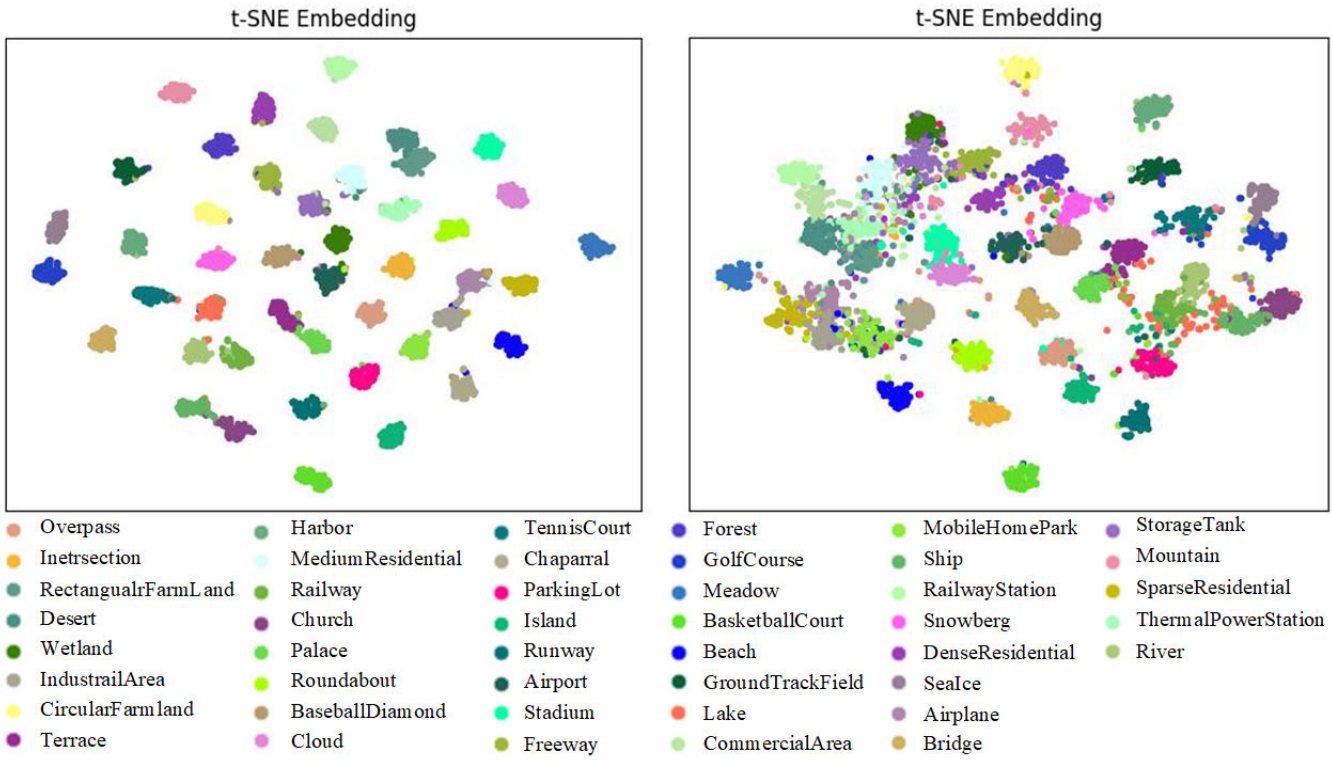

Fig. 7. T-SNE for NWPU-LT with an imbalanced ratio of 0.01 on (left) DECOR+ and (right) baseline.

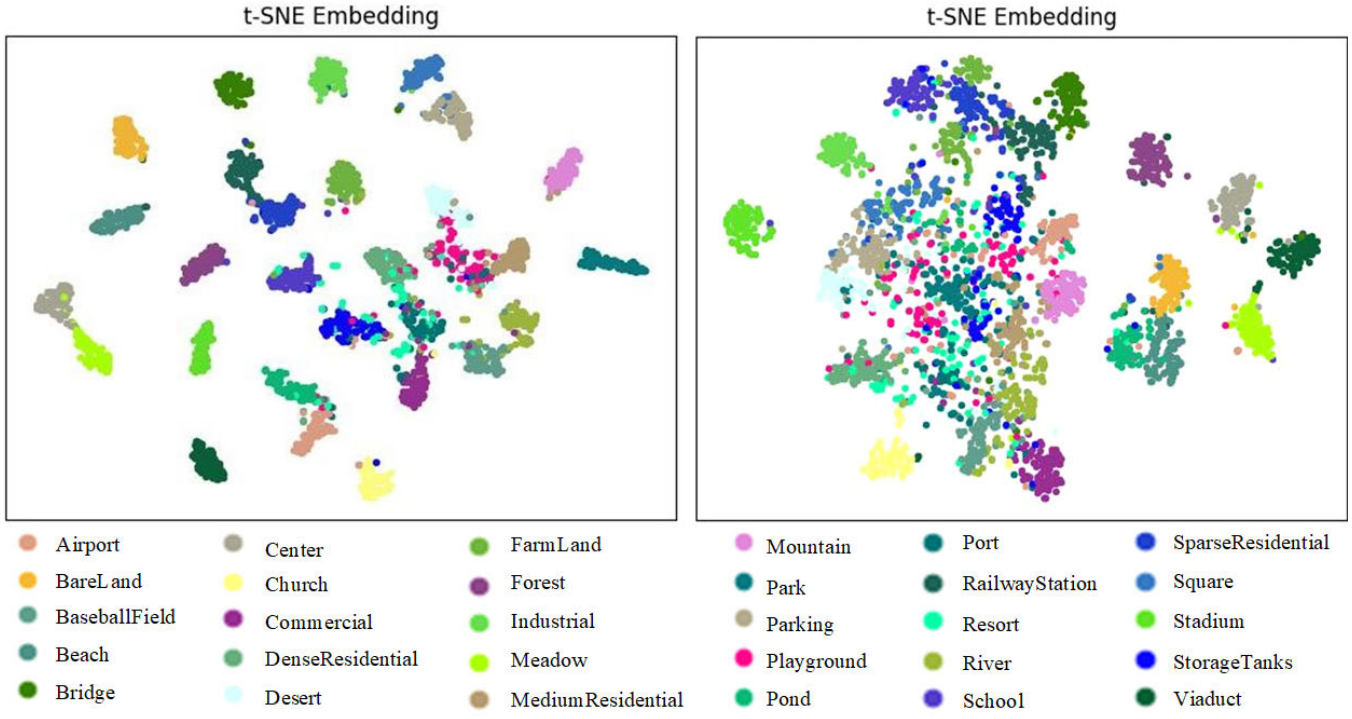

Fig. 8. T-SNE for AID-LT with an imbalanced ratio of 0.01 on (left) DECOR+ and (right) baseline.

recognition performance but also delivers pronounced benefits for tail classes—those that are typically the most difficult to learn under long-tailed distributions.

To further examine the spatial perception ability of the proposed method, we visualize the attention responses produced by the baseline model and DECOR+, as shown in Fig. 11. The enhanced network by DECOR+ consistently exhibits more complete and semantically aligned activation patterns. For instance, in Fig. 11(a), DECOR+ successfully highlights all four airplanes within the scene, whereas the baseline only focuses on three. A similar phenomenon is observed in Fig. 11(b), where the baseline emphasizes only the frontal region of the ship, while DECOR+ captures the entire structure. In addition, for elongated or line-shaped targets, such

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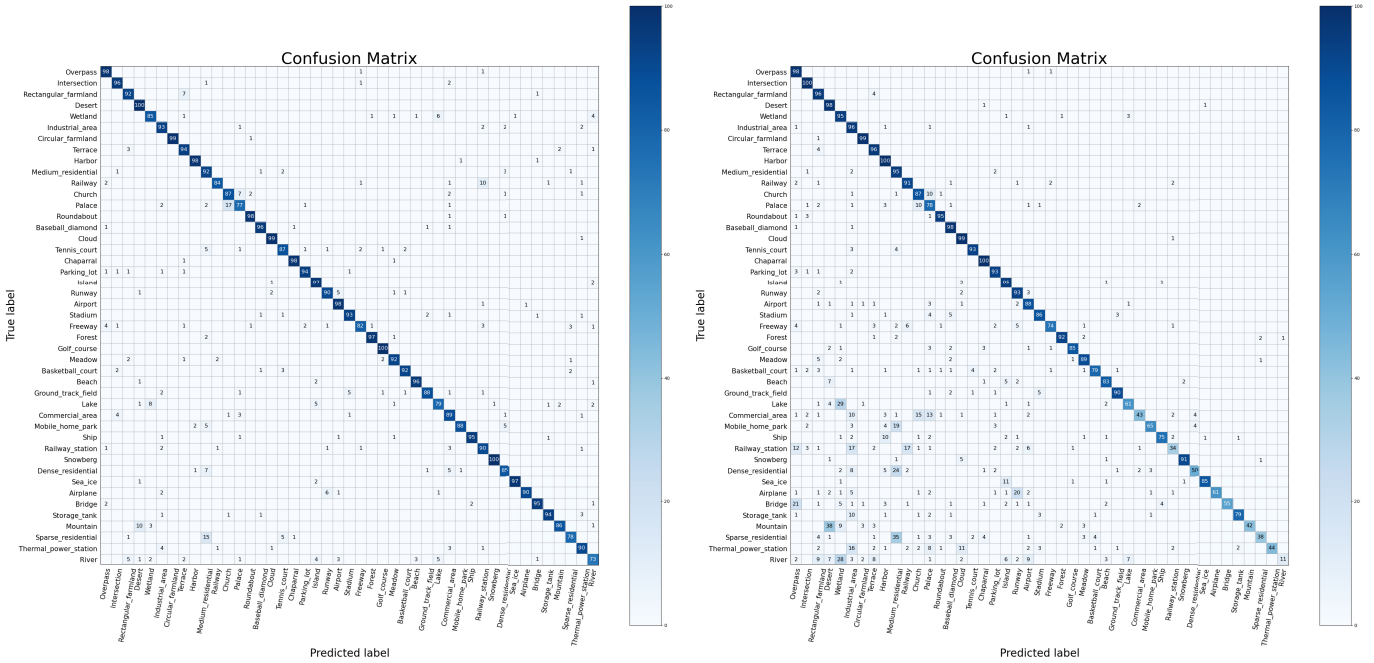

Fig. 9. Confusion matrix for NWPU-LT with imbalance ratio 0.01: (a) DECOR+ and (b) baseline.

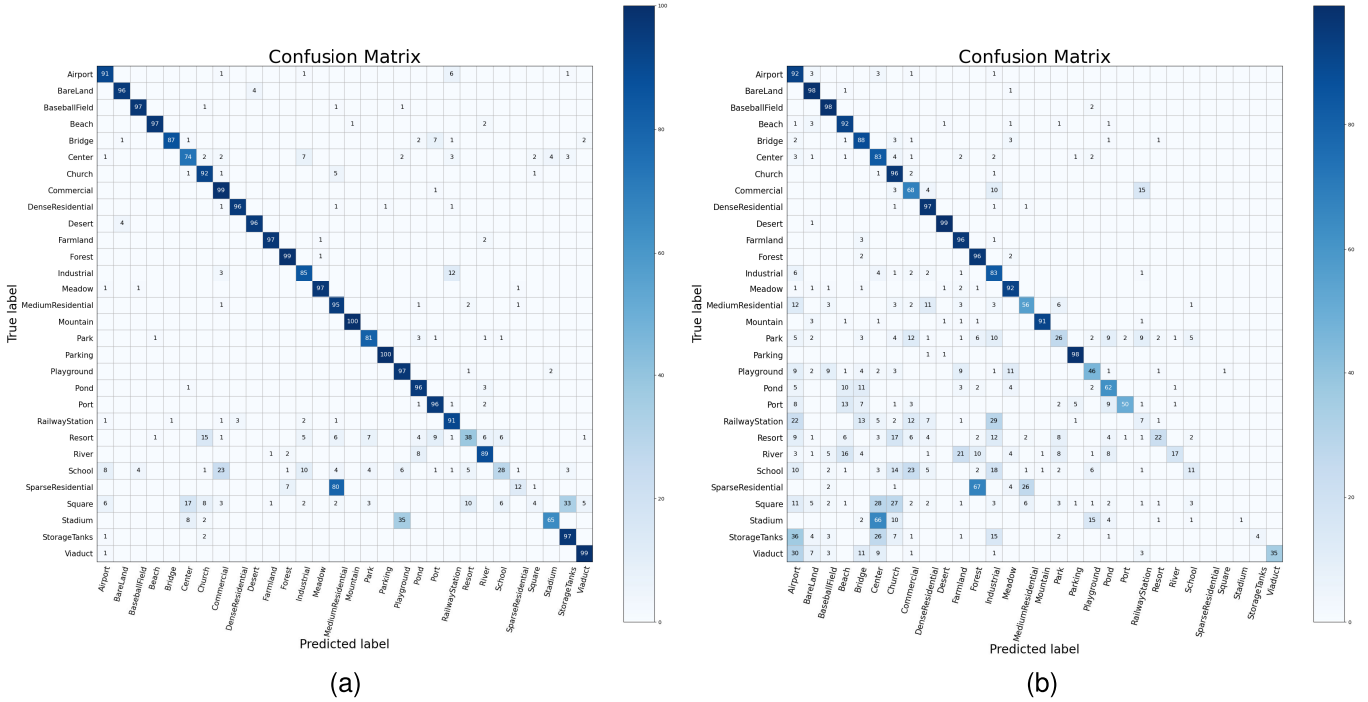

Fig. 10. Confusion matrix for AID-LT with imbalance ratio 0.01: (a) DECOR+ and (b) baseline.

as Fig. 11(c)–(e), our method is able to maintain a continuous attention trajectory across the full extent of the object, whereas the baseline often responds only to fragmented portions. These observations suggest that DECOR+ facilitates a more coherent spatial encoding, thereby yielding stronger representational robustness.

Beyond classification performance, the DECOR+ module is designed with architectural flexibility in mind. Due to its lightweight and modular nature, the method can be readily integrated into different backbone networks and transferred across diverse visual tasks. This adaptability makes DECOR+ a suitable candidate not only for long-tailed scene recognition but also for more structure-intensive tasks such as detection and segmentation. In ongoing and future work, we aim to incorporate DECOR+ into these domains to further investigate its benefits. Nevertheless, we acknowledge that the current version exhibits limited capability in handling completely unseen categories in zero-shot scenarios, which remains an important direction for future exploration.

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TABLE VII
COMPONENTWISE ABLATION EXPERIMENTAL RESULTS (%) ON NWPU-LT WITH IMBALANCED RATIO OF 0.01

Method	Module			Head	Middle	Tail	ALL
	LGF (M1)	PDP-SupCon (M2)	GTM (M3)				
Baseline(ResNet50)				98.00	94.56	68.77	79.88
M1	✓			97.67	92.94	72.42	81.40(+1.52)
M2		✓		97.33	94.63	84.00	88.67(+8.89)
M3			✓	97.33	92.56	72.96	81.56(+1.68)
M1+M2	✓	✓		96.33	92.25	88.46	90.33(+10.45)
M2+M3		✓	✓	95.67	92.69	88.96	90.73(+10.85)
M1+M2+M3	✓	✓	✓	95.33	92.56	90.54	91.58(+11.70)

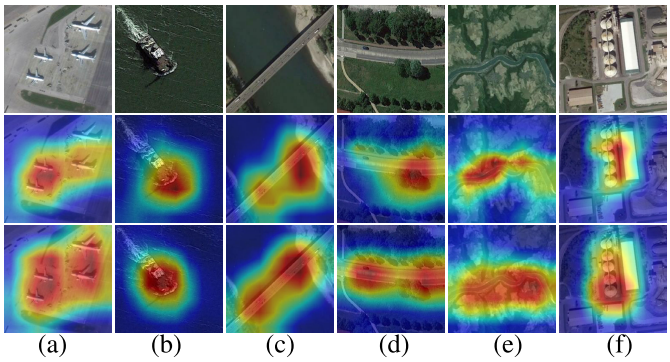

Fig. 11. Visualization of attention maps of (middle) baseline and (bottom) our DECOR+ model. (a) Airplane. (b) Ship. (c) Bridge. (d) Overpass. (e) River. (f) Storage tank.

E. Ablation Studies

This section presents a systematic ablation analysis to examine the effectiveness of each component in the proposed DECOR+ framework. All experiments are conducted on the ResNet-50 backbone, and the results are reported for the head, middle, tail, and ALL groups to reflect performance under long-tailed distributions.

1) Componentwise Analysis: We conduct a component-level ablation study to investigate the contribution of each module within the proposed DECOR+ framework. As reported in Table VII, ResNet-50 [50] is adopted as the reference backbone. Based on this baseline, *LGF* corresponds to the introduction of the LGF for frequency-domain enhancement, *PDP-SupCon* denotes the proposed polar dual-prototype supervised contrastive learning strategy, and *GTM* represents the GTM mechanism.

Starting from the baseline configuration, these components are progressively incorporated to form different model variants. Performance is evaluated across the head, middle, tail, and ALL category splits to reflect the characteristics of long-tailed distributions. The quantitative results of this incremental evaluation are summarized in Table VII.

2) Baseline Performance: For reference, the unmodified ResNet-50 baseline attains accuracies of 98.00%, 94.56%, and 68.77% on the head, middle, and tail groups, respectively, with an overall accuracy of 79.88%. This baseline configuration establishes the performance level prior to incorporating any DECOR+ components and provides a stable foundation for subsequent comparisons.

3) Effect of the LGF: Introducing the LGF module leads to noticeable improvements. The accuracy of head and medium classes slightly decreased, while the tail class exhibited a substantial increase, reaching 72.42% (+ 3.65%). The overall accuracy reaches 81.40% (+ 1.52%), revealing that the frequency-domain global enhancement substantially strengthens the model's ability to capture discriminative signals for low-frequency and visually sparse categories.

4) Impact of the PDP-SupCon Learning: Introducing the PDP-SupCon module leads to the most significant improvement among the three individual components. The accuracies of head and medium classes remain largely stable, while the tail class shows a substantial increase, reaching 84.00% (+ 15.23%). The overall accuracy reaches 88.67% (+ 8.89%), indicating that the proposed dual-prototype supervised contrastive learning effectively enhances the discriminative structure of the embedding space and provides much stronger representation support for tail classes under long-tailed distributions.

5) Influence of the GTM: Introducing the GTM module also brings consistent performance gains. Although the accuracies of head and medium classes slightly decrease, the tail class improves noticeably to 72.96% (+ 4.19%). The overall accuracy reaches 81.56% (+ 1.68%), showing that the Gaussian-based tail modeling strategy can effectively improve the statistical representation of underrepresented classes and alleviate the optimization bias toward head classes, thereby enhancing recognition performance for tail categories.

6) Combined Influence of LGF and PDP-SupCon: Augmenting LGF with the PDP-SupCon objective results in further structural refinement of the representation space. While the head classes experience a slight drop to 96.33%, the tail classes achieve a substantial improvement, reaching 88.46%. The overall accuracy increases to 90.33% (+ 10.45%), indicating that the contrastive refinement encourages more balanced class separation, particularly enhancing minority-class discrimination with only a negligible cost to the head classes.

7) Complementary Effect of PDP-SupCon and GTM: When GTM is combined with PDP-SupCon, the tail category performance rises to 88.96%, and the overall accuracy increases to 90.73%. This confirms the role of GTM in further strengthening the representation quality of underrepresented categories by explicitly enhancing the tail class embedding distribution.

8) Joint Contribution of LGF, PDP-SupCon, and GTM: Finally, integrating all three components yields the most balanced performance profile. Compared with the LGF + PDP-SupCon configuration, the head and middle categories slightly decreased, while the tail category exhibits additional gains,

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TABLE VIII
EFFECT OF THE RADIAL TERM WEIGHT λ ON NWPU-LT (%) WITH AN IMBALANCE RATIO OF 0.01

λ	Head	Middle	Tail	ALL
0.5	94.33	93.25	89.96	91.42
1	95.33	92.56	90.54	91.58
1.5	95.00	92.94	90.19	91.49
2	96.67	93.13	89.81	91.44
2.5	96.00	92.81	89.85	91.31

with the tail class reaching 90.54% (+ 2.08%). Relative to the PDPSupCon + GTM configuration, DECOR+ with all components achieves improvements of + 1.58% in the tail category, and raises the overall performance by + 0.85%. These results demonstrate that the three modules reinforce each other. LGF enhances global feature expressiveness, PDP-SupCon encourages uniform class separation, and GTM improves tail-specific representational density, together forming a cohesive and highly effective long-tailed recognition framework.

9) Effectiveness of the Radial Energy Regulation Weight λ: To evaluate the sensitivity of PDP-SupCon to the weighting of the radial contrastive loss λ (which balances the angular and radial contrastive terms), we conduct an ablation study on NWPU-LT under an imbalance ratio of 0.01. Specifically, we vary λ from 0.5 to 2.5 while keeping all other settings unchanged. The results in Table VIII show that the proposed method exhibits low sensitivity to λ within this range: the overall accuracy (ALL) remains stable (91.31%–91.58%, δ = 0.27%), and the tail-class accuracy varies only slightly (89.81%–90.54%, δ = 0.73%). Based on this ablation, we set λ = 1 as the default choice, as it achieves the best overall accuracy and provides a favorable balance across head/middle/tail performance.

V. CONCLUSION

This article presents DECOR+, a unified energy-balancing framework for long-tailed RSIC. By jointly enhancing deep representations in the spectral domain, restructuring embedding geometry through prototype-guided contrastive learning, and rebalancing optimization dynamics via Gaussian-based tail modeling, DECOR+ effectively mitigates the intertwined challenges of feature degradation, distribution skewness, and gradient dominance that commonly arise under long-tailed conditions. Extensive experimental results on multiple long-tailed RS benchmarks demonstrate that the proposed framework consistently improves both overall recognition accuracy and tail-class performance across varying imbalance severities while maintaining robustness and architectural flexibility. Beyond empirical gains, this work highlights the importance of addressing long-tailed recognition from a holistic perspective, where feature quality, embedding distribution, and decision boundary fairness are jointly considered rather than optimized in isolation. Due to its lightweight and modular design, DECOR+ can be seamlessly integrated into existing backbone architectures without structural modifications. Future research will explore extending this framework toward more challenging open-set and open-world RS scenarios, as well as adapting the proposed energy-balancing principles to other vision tasks, such as object detection and semantic segmentation, further broadening its applicability in real-world long-tailed environments.

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